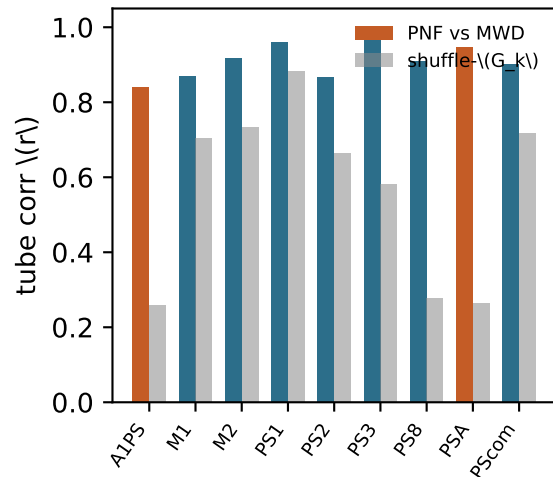
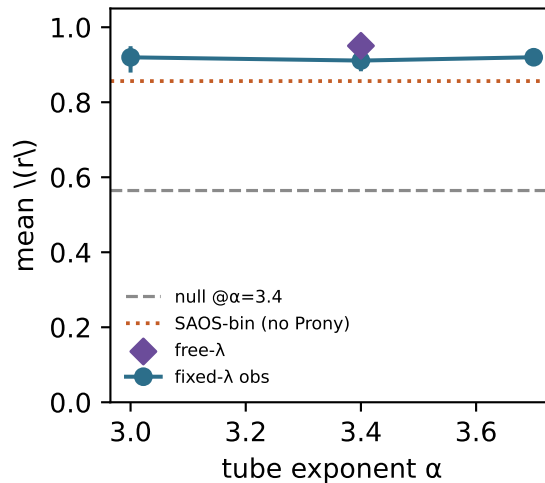


(a) Leeds tube alignment



(b) Robustness controls

**(c) Locked numbers**

Leeds n=9 harden

obs $r = 0.911$

CI [0.88, 0.94]

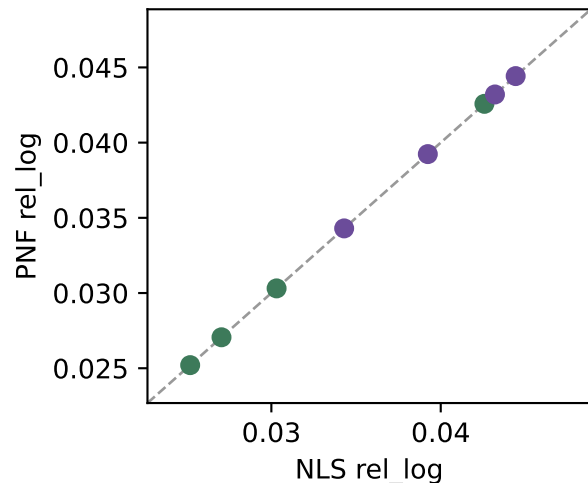
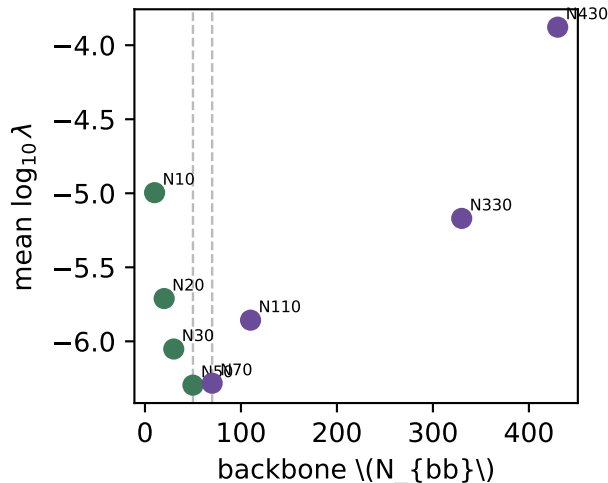
null $r = 0.565$

$\Delta = 0.347$

frac $p < 0.05 = 0.89$

SAOS-bin $r = 0.856$

free- λ $r = 0.950$

(d) UMN PNF $\equiv$ NLS(e) Spectrum vs topology $\rho=0.19$ **(f) Top-end discovery gates**

NOW (publishable core)

- Mode mass $\gg$ shuffle-G null ($\Delta \approx 0.35$)
- α -robust; free- λ preserved
- PNF $\equiv$ NLS on PS + PLA grafts

CEILINGS (keep in paper)

- Abs. r partly structural (bin ≈ 0.86)
- $n=9$; LOO Mw fails; Elliott wins MWD
- UMN: no $w(M)$; no $N_{\text{bb}} \propto \log \lambda$ link

NATURE GATES

- Unique map / larger paired n
- Cross-chemistry nontrivial predict
- Close gap to nonlinear tube / NN